

Factorial Experiments

An experiment in which all levels of a given factor are combined with all levels of every other factor in the experiment is called a *factorial experiment*. An important issue to be examined in a factorial experiment is how the levels of various factors interact with each other in their effect on the response variable. When the differences in the response variable between the levels of one factor are not the same at all levels of the other factors, we say that an *interaction* between the factors is present. To model and study this interaction we will need at least two observations for each combination of the levels of the factors in the experiment.

Two Factor Factorial Experiment (fixed effects)

The Castle Bakery company supplies wrapped Italian bread to a large number of supermarkets in a metropolitan area. An experimental study was made of the effects of height of the shelf display (bottom, middle, top) and width of the shelf display (regular, wide) on sales of this bakery's bread (measured in cases) during the experimental period.

Factor B (display width) j

Factor A (display height) I	B ₁ (regular)	B ₂ (wide)
A ₁ (bottom)	47	46
	43	40
A ₂ (middle)	62	67
	68	71
A ₃ (top)	41	42
	39	46

Model: $Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{k(ij)},$

$$i=1,\dots,a, j=1,\dots,b, k=1,\dots,n.$$

where

$$E(Y_{ijk}) = \mu_{ij} = \mu + A_i + B_j + (AB)_{ij}$$

and

$$A_i + B_j + (AB)_{ij}$$

are the mean and the effect of the ij th treatment, respectively.

Assumptions:

$$1. \sum_i A_i = 0$$

$$2. \sum_j B_j = 0$$

$$3. \sum_i (AB)_{ij} = 0$$

$$4. \sum_j (AB)_{ij} = 0$$

$$5. \varepsilon_{k(ij)} \text{ are iid } N(0, \sigma_e^2).$$

We will examine a SAS output that presents the statistical data analysis, examines the underlying assumptions for the model for this data and presents a plot for the interactions between the two factors.